

Design and Implementation of an 8×4 SRAM Array with Peripheral Circuits in 90nm CMOS

Tools Used

- Cadence Virtuoso
- Assura DRC/LVS
- Spectre Simulator

Technology

- GPDK 90nm CMOS

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Introduction

Static Random Access Memory (SRAM) is a high-speed semiconductor memory widely used in processors, cache memories, and embedded systems. Unlike DRAM, SRAM does not require periodic refreshing and offers faster access times, making it suitable for performance-critical applications.

A complete SRAM architecture consists of memory cells along with peripheral circuits such as decoders, precharge circuitry, and sense amplifiers. These peripheral blocks play a vital role in ensuring reliable read and write operations.

This project presents the design, simulation, and layout implementation of an 8x4 SRAM array using custom 6T SRAM bitcells in GPDK 90nm technology.

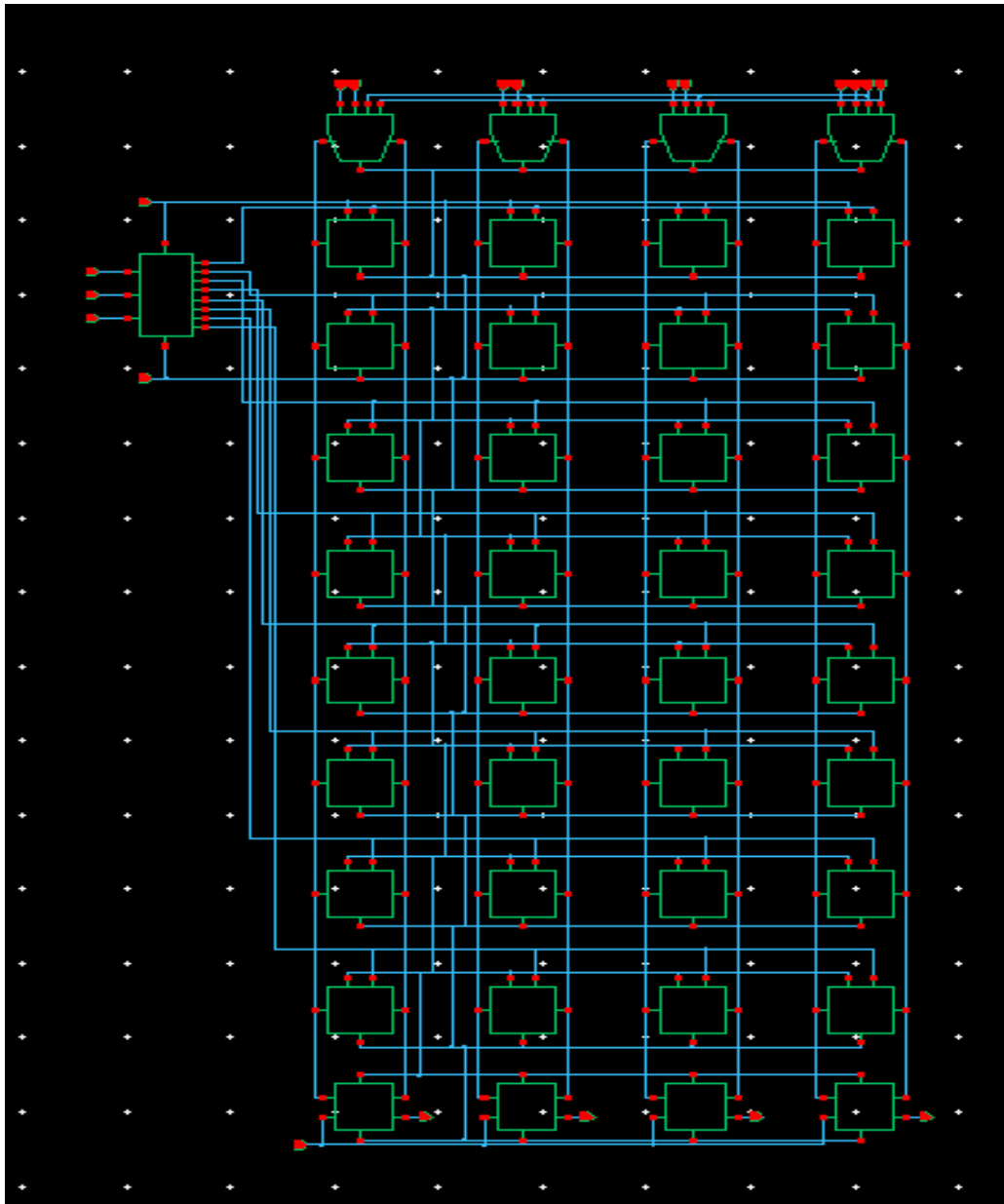


Fig. 1: Schematic of 8x4 6T SRAM Array with Peripheral Circuits

6T SRAM Bitcell

The memory array is constructed using a conventional six-transistor SRAM cell consisting of:

- Two cross-coupled CMOS inverters
- Two access transistors
- Differential bitlines (BL and BLB)
- Wordline controlled access mechanism

Cell Features

- Differential read operation
- Differential write operation
- Static data retention capability
- Low transistor count
- High integration density

Cell Parameters

Parameter	Value
Technology	90nm
Cell Type	6T SRAM
Cell Area	7.88 μm^2
Hold SNM	180 mV

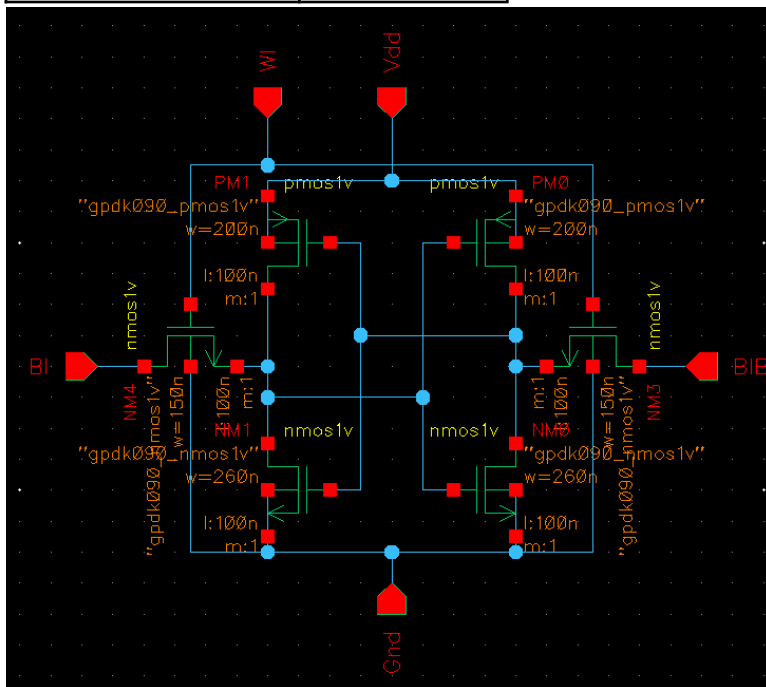


Fig. 2: Schematic of 6T SRAM Cell

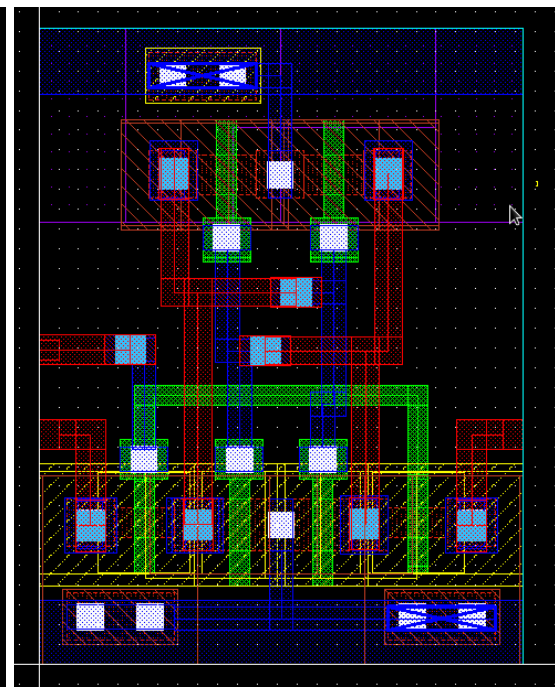


Fig. 3: Layout of 6T SRAM Cell with clean DRC and LVS

Peripheral Circuit Implementation

To enable complete memory functionality, several supporting circuits were designed.

Row Decoder

The decoder converts the address inputs into unique wordline activation signals. Only one row is selected at a time during memory access.

Precharge Circuit

The precharge circuit equalizes and charges both bitlines before every read operation, ensuring reliable sensing.

Sense Amplifier

A differential digital sense amplifier was designed to detect small voltage differences between BL and BLB and convert them into full-swing digital outputs.

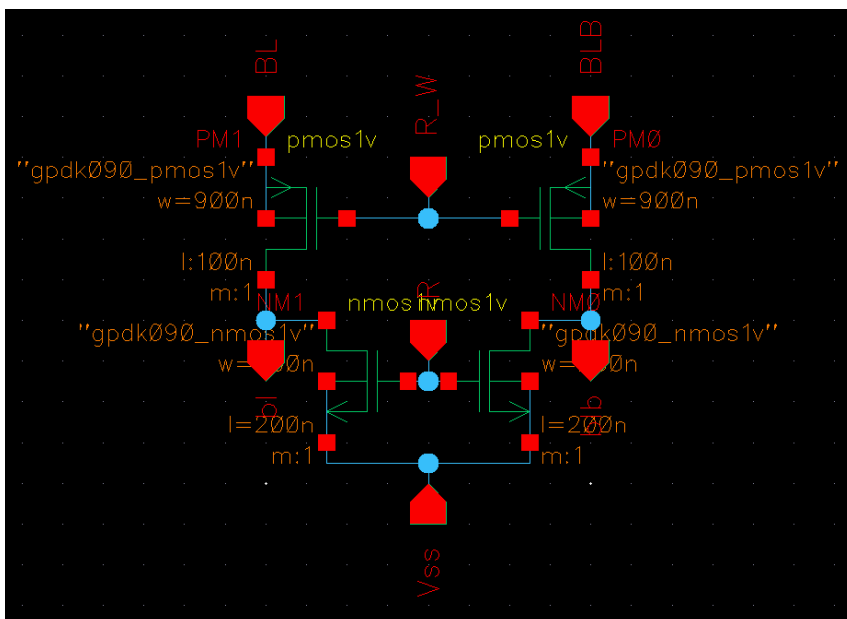


Fig. 4: Schematic of Pre-Charge Circuit

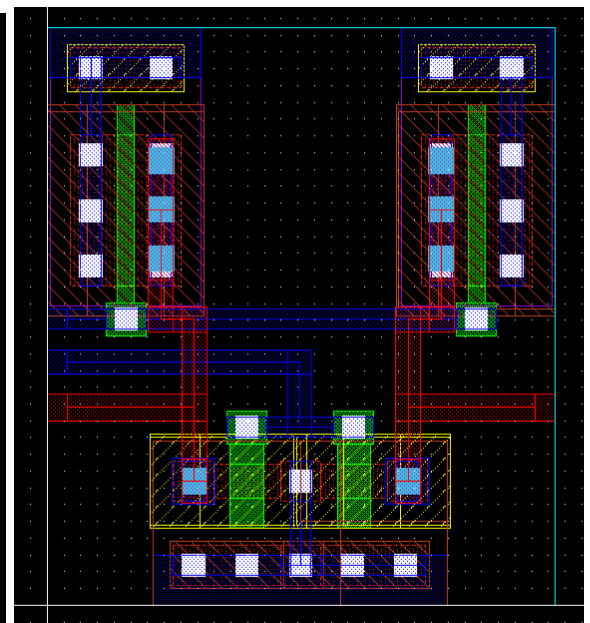


Fig. 5: Layout of Pre-Charge Circuit with clean DRC and LVS

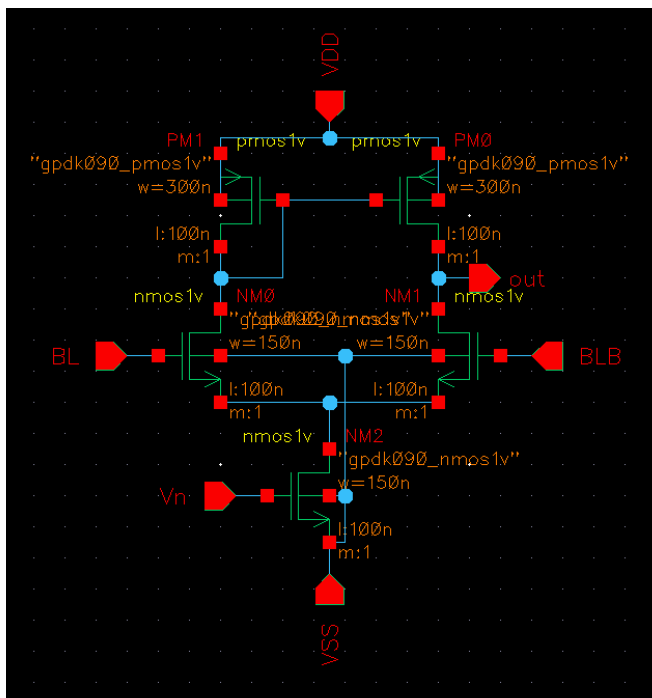


Fig. 6: Schematic of Sense Amplifier Circuit

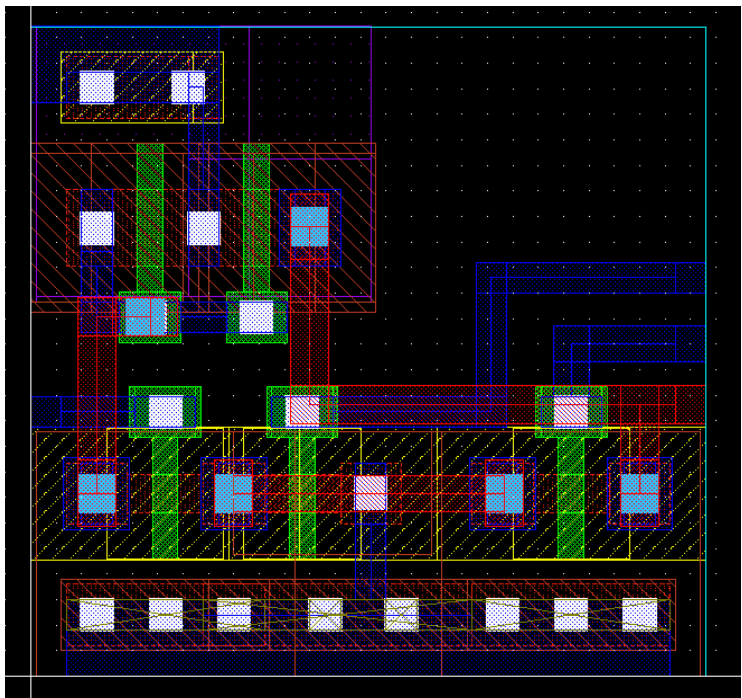


Fig. 7: Layout of Sense Amplifier Circuit with clean DRC and LVS

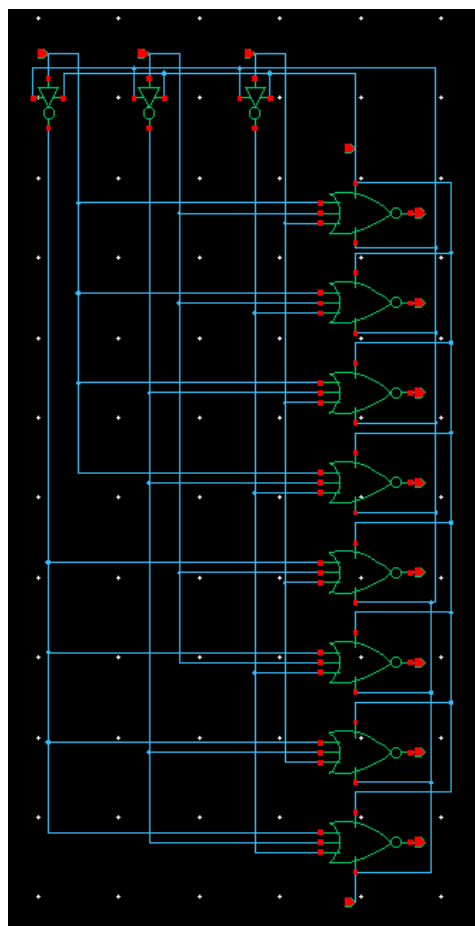


Fig. 8: Schematic of Decoder Circuit

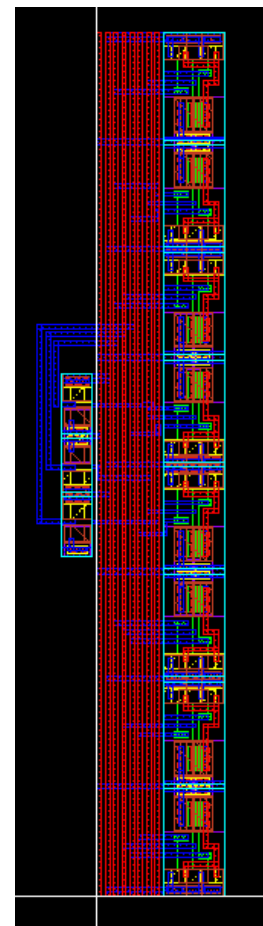


Fig. 9: Layout of Decoder Circuit with clean DRC and LVS

SRAM Array Organization

An 8×4 memory architecture was implemented using custom-designed 6T SRAM cells.

Architecture Features

- 8 memory rows
- 4-bit word width
- Shared bitline architecture
- Decoder-based row selection
- Differential sensing mechanism

The array integrates memory cells and peripheral circuits into a complete SRAM subsystem capable of performing read and write operations.

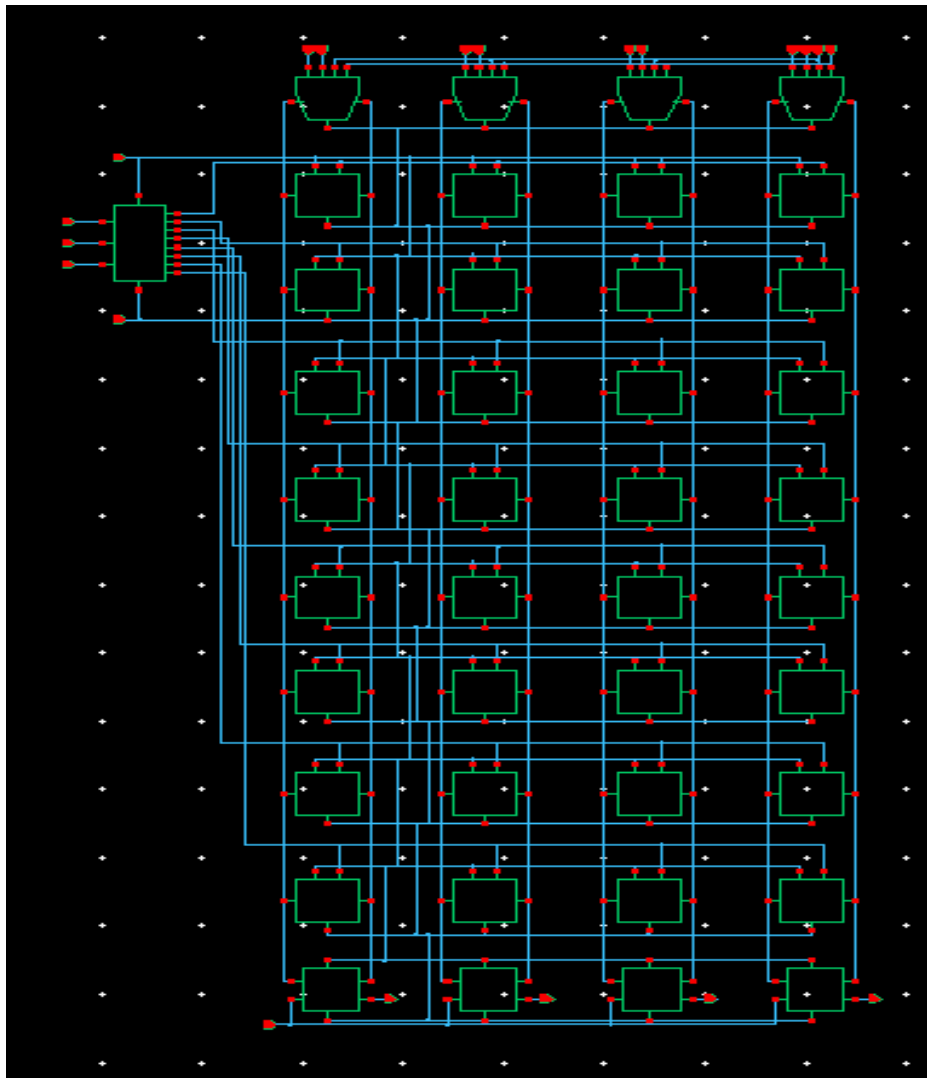


Fig. 10: Schematic of 8x4 6T SRAM Array with Peripheral Circuits

Read and Write Operation Verification

Transient simulations were performed to verify memory functionality.

Write Operation

- Address decoded successfully
- Wordline activated
- Data written through differential bitlines
- Data retained after wordline deactivation

Read Operation

- Bitlines precharged
- Selected cell accessed
- Differential voltage generated
- Sense amplifier detected stored data correctly



Fig. 11: Full Waveform SRAM Array with Peripheral Circuits verify Read, Write and Hold operation within Array

Performance Analysis

The SRAM cell and array were characterized using transient simulations.

Parameter	Value
Read Delay	5.21 ps
Write '0' Delay	16.09 ps
Write '1' Delay	26.23 ps
Hold SNM	180 mV
Cell Area	7.88 μm^2

Observations

- Successful read and write operations achieved.
- Stable data retention observed.
- Peripheral circuits operated correctly.
- Compact memory implementation achieved.

Static Noise Margin (SNM)

The stability of the SRAM cell was evaluated using butterfly curve analysis.

The Hold Static Noise Margin (SNM) was measured to be:

$$\text{Hold SNM} = 180 \text{ mV}$$

This value indicates reliable data retention and robustness against noise disturbances during hold mode operation.

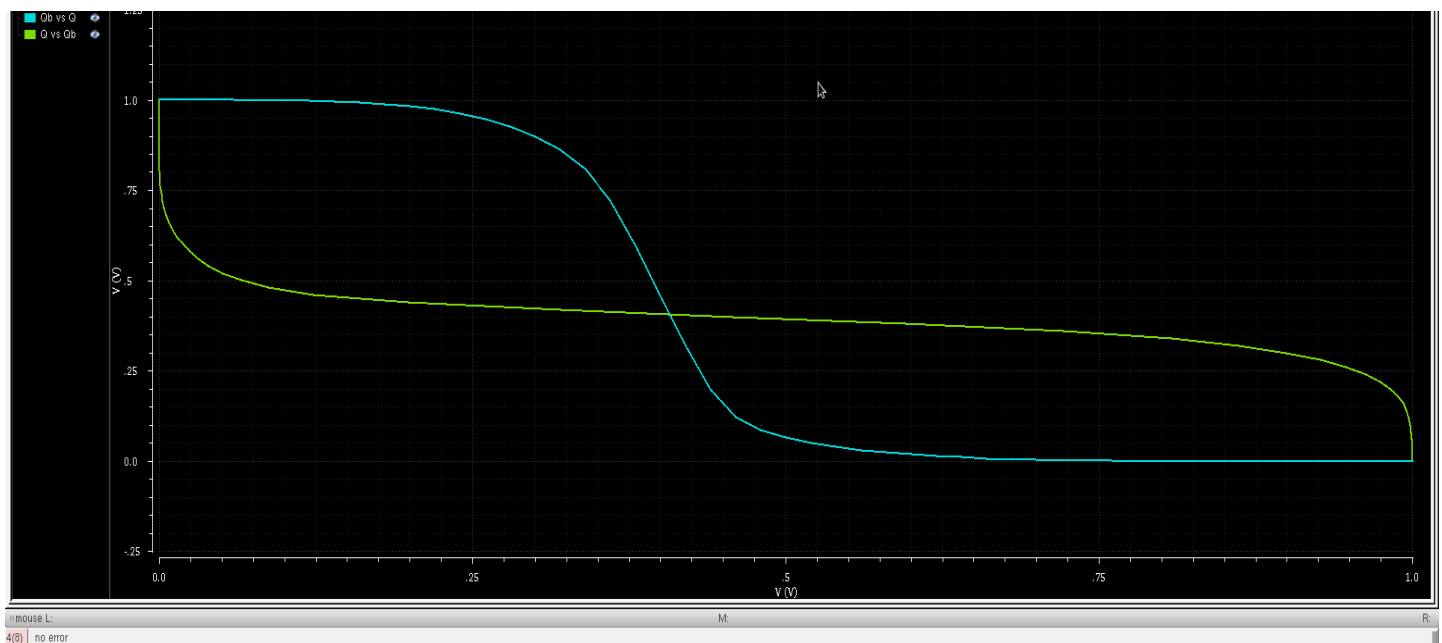


Fig. 12: Butterfly Curve of 6T SRAM Cell

Layout Verification

All layouts were verified using industry-standard physical verification checks.

DRC Verification

- No design rule violations detected.
- Layout satisfies all GPDK90 design constraints.

LVS Verification

- Layout matches schematic.
- No device mismatches.
- No net mismatches.
- No pin mismatches.

Verification Summary

Check	Result
DRC	Passed
LVS	Passed
Device Mismatch	0
Net Mismatch	0

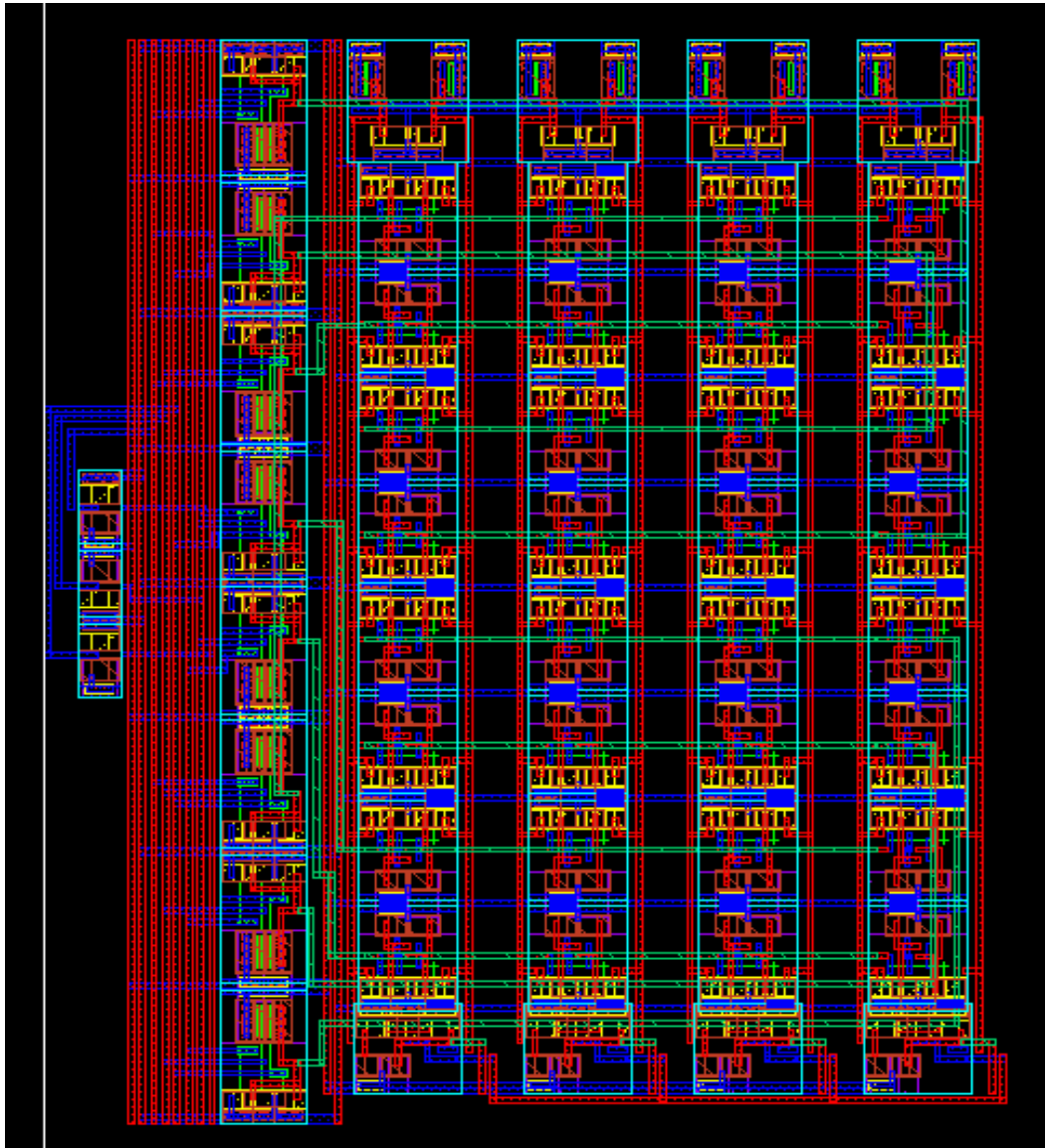


Fig. 13: Layout of 8x4 6T SRAM Array with Peripheral Circuits

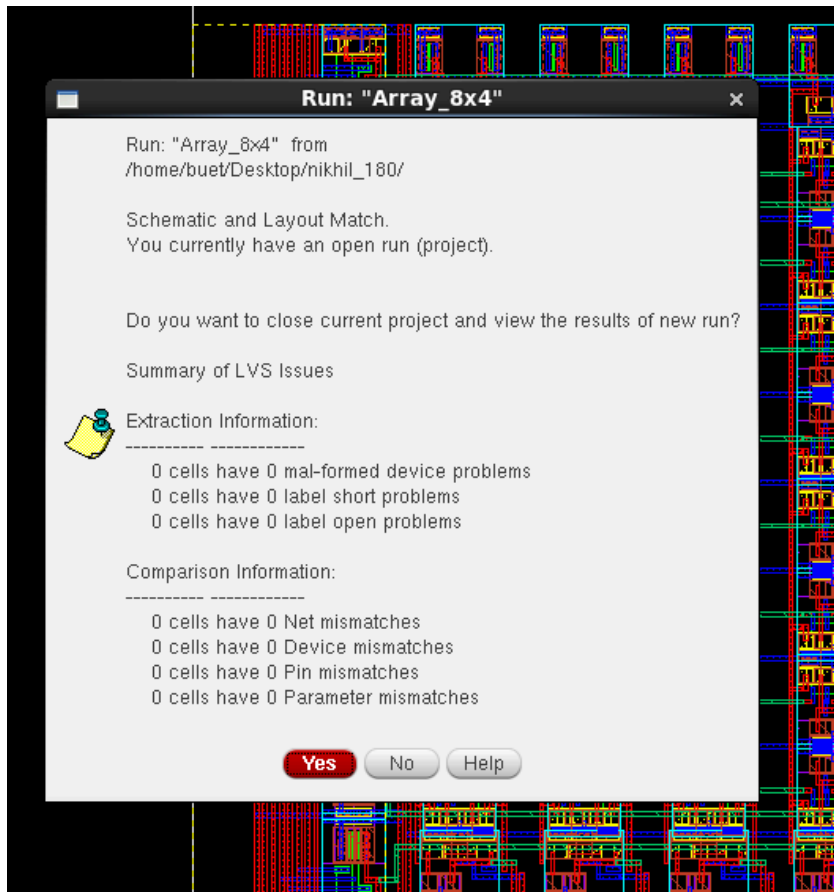


Fig. 14: Clean LVS of 8x4 SRAM Array

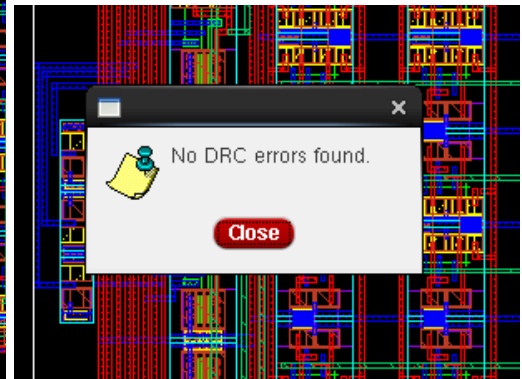


Fig. 15: Clean DRC of 8x4 SRAM Array

Conclusion

A complete 8×4 SRAM array was successfully designed and implemented in GPDK 90nm CMOS technology using Cadence Virtuoso.

Key Achievements

- ✓ Custom 6T SRAM Cell Design
- ✓ Row Decoder Design
- ✓ Precharge Circuit Design
- ✓ Differential Sense Amplifier Design
- ✓ 8×4 SRAM Array Integration
- ✓ Functional Read/Write Verification
- ✓ Hold SNM Analysis
- ✓ Full-Custom Layout Implementation
- ✓ Clean DRC Verification
- ✓ Clean LVS Verification

This project provided practical experience in memory architecture design, SRAM characterization, peripheral circuit integration, custom layout implementation, and physical verification flow.